Solution overview: Credit Risk Portfolio Monitor

14,685 loan accounts from 5 synthetic bank tables · Streamlit dashboard with two embedded AI features · Python, pandas, scikit-learn, LightGBM

1. What the dashboard shows and how an analyst uses it

- **Overview.** KPIs, exposure and observed vs predicted default by model risk band, trends and profiles by income, employment, region, age and loan type, a cross-dimension heatmap and the portfolio insights. Start here to see where risk and money sit.
- **Segments & drill-down.** Choose a dimension and a segment to list its accounts ranked by predicted risk; open any account or download the list.
- **Account.** Predicted default probability, risk band, the top drivers of the score (TreeSHAP), 12-month repayment chart, bureau lines, peer comparison and any anomaly reasons.
- **Ask the Portfolio.** Plain-English questions answered with exact figures and their source (AI feature 1).
- **Anomaly review.** The review list with reasons and a Confirm/Dismiss decision log (AI feature 2).
- **Model** (ROC, KS, SHAP, calibration, PSI monitoring) and **Data & methods** (join logic, data-quality log, feature cutoffs).
- Sidebar filters (loan type, employment, region, income, age, risk band, origination year) re-cut every portfolio view.

2. The two AI features

Ask the Portfolio (LLM)	**Anomaly detection**
How. An analyst types a question. Guardrails run first. An LLM (or the built-in offline interpreter) turns the question into a small JSON query plan from a closed list of metrics, dimensions and filters. pandas runs the plan locally and the answer is written from the executed result, so every number in the answer comes from the data.	**How.** Each account is compared with its peer segment (same loan type and income band) and with its own history (last 3 months versus first 9). A signal fires when a value is beyond the peer group's 99th percentile and above a materiality floor, or when the account's own payment behaviour breaks sharply. Accounts with two or more signals, or a sharp own-history break, are flagged.
Trigger. Runs when an analyst submits a question. Questions about forecasts, causes, individual accounts, raw rows, credentials or anything outside the five tables are declined before any planner sees them.	**Trigger.** Runs on every data build over all accounts. An account is flagged when at least two independent signals fire, or when its days past due jump by 25+ days (to 30+) or its paid ratio drops by 50+ points between months 1-9 and months 10-12.
False positive. A confident but wrong answer: the planner reads the question differently from the analyst (for example 'risky' taken as default rate, or the wrong segment), or a question that should have been declined gets answered. The exact plan and query are shown with every answer so this can be spotted.	**False positive.** A flagged account that is actually fine: a one-off payment pause, a bank processing error, a customer with many legitimate credit lines, or an unusual but healthy account in a small peer group. The analyst dismisses it with a note.
False negative. A valid question that is declined or only partly understood (for example 'more than 2 missed payments', which is not a supported filter), or the free-tier LLM being rate limited so the offline interpreter handles it. Cost: the analyst rephrases or picks an example question.	**False negative.** A risky account that is not flagged: slow deterioration that never crosses the 99th-percentile or two-signal bar, or an account with no bureau file (bureau signals cannot fire). The model risk score is the second line of defence.
Human review. The analyst reads the answer and the 'Where this came from' panel (metric, filters, grouping, source) before using a figure. Nothing is written back to any system, and the LLM never sees row-level data or results. **Sent to LLM:** Only the question text and the table catalog (metric and column descriptions plus category names) are sent to the LLM. No rows, no computed results, no identifiers.	**Human review.** The analyst opens each flagged account, reads the reasons with their numbers and records Confirm or Dismiss with a note. Each decision is stored with timestamp, account, decision, note and reviewer. Decisions never change the model or the account. Result: 238 flagged (1.6%); flagged accounts default at 30% vs 16%.

3. Model results and approach

Model (test set)	AUC (95% CI)	KS	Gini
Logistic regression (baseline)	0.683 (0.654-0.708)	0.268	0.366
LightGBM (challenger)	0.678 (0.650-0.704)	0.279	0.356

Test set: 2,948 accounts (16.0% default), split by customer so no customer appears in both sets. 49 features from loan, customer, repayment months 1-12 and bureau data; collections data and protected attributes are excluded. Tuning used GroupKFold on training rows only; the test set was scored once. **No reliable difference between the models: LightGBM minus logistic AUC = -0.005 (95% paired-bootstrap CI -0.014 to 0.003 includes 0). The simpler, more explainable model is a defensible choice.** Dashboard scores use Logistic regression, chosen by cross-validated AUC. Robustness: profile-only AUC 0.645, plus repayment 0.673, plus bureau 0.678; bureau limited to lines opened by month 12: 0.674. Strongest single feature AUC 0.618 (leakage ceiling 0.95).
Monitoring: Population drift: Weekly PSI on the score and on the top-10 features vs the training sample. PSI < 0.10 stable, 0.10-0.25 watch, > 0.25 investigate. Discrimination: Monthly AUC / KS / Gini on matured cohorts (accounts with a known outcome); alert if AUC falls more than 0.03 below the validation value or the paired-bootstrap interval excludes it. Calibration: Compare mean predicted PD with observed default rate by risk band each month; alert if any band is off by more than 20 % relative. Data quality: Daily checks on null rates, new category values, bureau extract freshness and the share of imputed fields (the DQ flags in this repo).

4. Top 3 portfolio insights

- **Mortgages are the riskiest product and also 52% of exposure.** Mortgage default rate is 24.7% vs 16.1% for the portfolio and 13.9% for Auto Loan. Because it carries 52% of exposure, a few points of extra default here matter more than any other product. Secured does not mean safe: review underwriting and collateral valuation on this book first.
- **Default risk falls steadily with income: 39% below 30k vs 8% above 120k.** Customers earning under 60k are 22% of accounts but 34% of defaults. Age follows the same pattern (22.4% under 25 vs 10.6% at 55+). Affordability checks and lower limits for low-income, young applicants are the cheapest lever.
- **A missed payment in the first 3 months lifts default risk from 15% to 22%.** 1,905 accounts (13.0%) missed at least one of their first three payments and default at 21.5% vs 15.3%. Accounts that never miss in 12 months default at 12.9%. An early-collections trigger at month 3 reaches the riskiest accounts while there is still time to restructure.

5. Key assumptions and what production needs from the bank

Assumptions. The observation point is the end of repayment month 12 and the target is assumed to be observed afterwards (no default date exists). The bureau extract has no as-of date, so it is used as supplied (sensitivity above). Exposure is the original loan amount. Medians used for imputation never use the target. Data is synthetic: results illustrate the method, not any real bank.
From IT and data teams. (1) Scheduled extracts from core banking, the repayment ledger with true balances, bureau pulls with an as-of date, and the collections system, under data contracts with lineage. (2) A dated default definition (for example 90+ days past due) and clear observation and outcome windows. (3) A secured environment: SSO and role-based access, PII masking, audit logging of every analyst decision, and an approved LLM gateway or an on-premise model so no customer data leaves the bank. (4) Case-management integration for anomaly decisions. (5) Model-risk validation, fairness review and the monitoring thresholds above.